

Experiment 1 report

Experiment 1-1

Contents:

Design a rounding discriminant circuit (its input is 8421 BCD code, when the input is greater than or equal to 5, the discriminant circuit output is 1, vice versa is 0. Design with only NAND gates).

Design process:

Firstly, we can draw the truth table.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1

Then, from the table, we can generate the logic expression of it.

$$F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$$

We can use k – map to simplify this expression.

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

After simplification, we can get:

$$F = A + BD + BC$$

Because we can only use NAND gate,

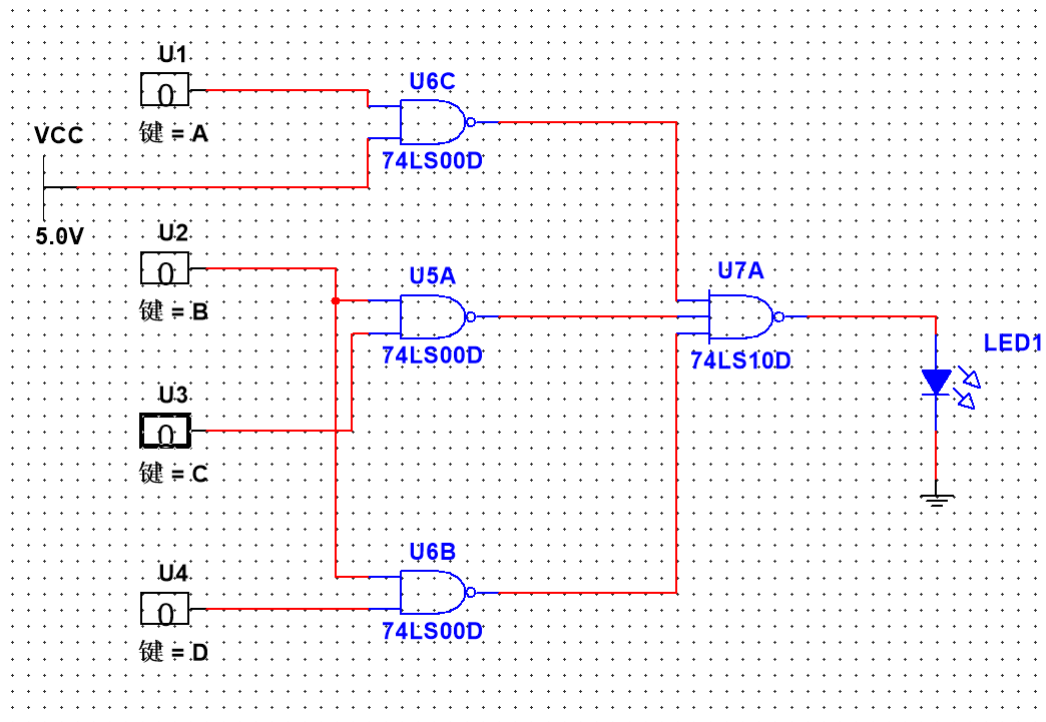
$$F = \overline{\overline{A + BD + BC}}$$

$$F = \overline{(A1)(BD)(BC)}$$

Now, we can use Multisim to simulate the circuit.

Result

The circuit diagram is as shown.



Discuss or comment

This circuit uses only NAND gates to implement the function that judges whether the input is greater than or equal to 5 or not. In this experiment, I used the truth table and K-map method to complete this task.

In the test case, when the input is 0101, 0110, 0111, 1000, 1001, the output is 1. When the input is 0000, 0001, 0010, 0011, 0100, the output is 0. This indicates that the circuit is correct.

(We use the LED light to show the output, on is 1, off is 0. We use the VCC to indicate a NAND with 1.)

Experiment 1-2

Contents

Design a voting circuit. 3 inputs are A,B and C. 1 control line M. When M = 0, if A,B, and C all agrees, F = 1,otherwise F = 0. When M = 1, if the majority of A,B and C agree , F = 1,otherwise F = 0. Design with 3-8 decoder 74LS138 and necessary gates.

Design process

Firstly, we draw the truth table.

M	A	B	C	F
0	1	1	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1
x	x	x	x	0

Then, we can generate the logic function.

$$F = \bar{M}ABC + M\bar{A}BC + MAB\bar{C} + MABC$$

We can use K-map to simplify the expression.

MA \ BC	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

$$F = \sum m(7,11,13,14,15)$$

$$F = \overline{m_7} + \overline{m_{11}} + \overline{m_{13}} + \overline{m_{14}} + \overline{m_{15}}$$

$$F = \overline{(m_7)(m_{11})(m_{13})(m_{14})(m_{15})}$$

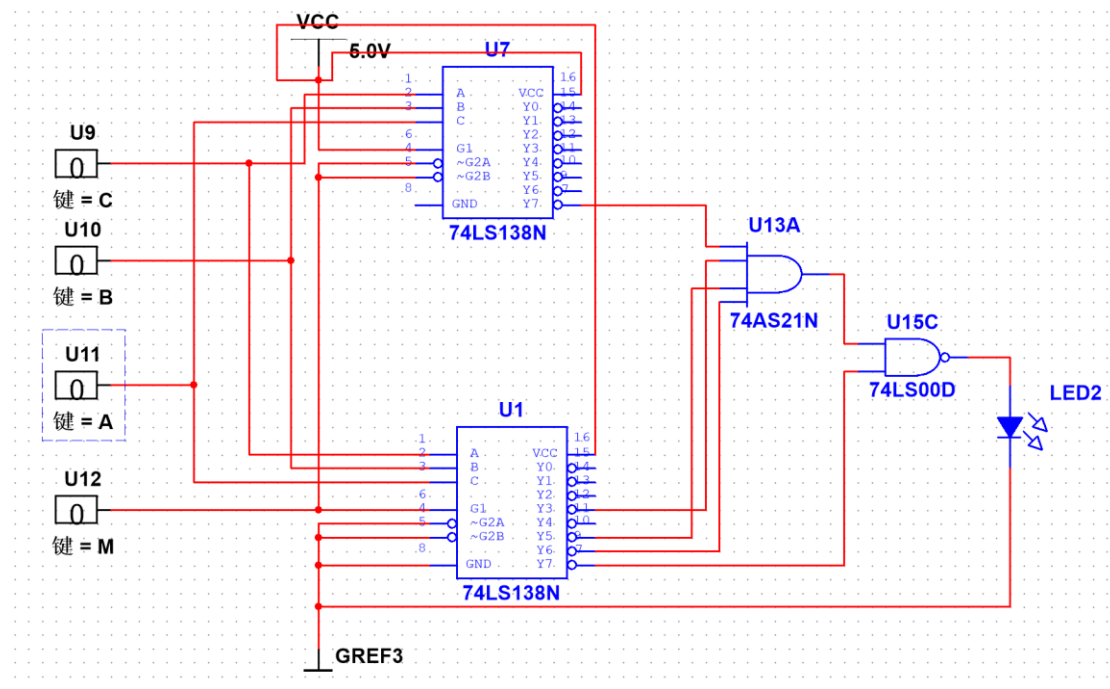
$$F = \overline{(F_7)(F_{11})(F_{13})(F_{14})(F_{15})}$$

Because there is 4 variables, so we can first expand the 3-8 decoder to 4-16 decoder.

We use 2 of the 3-8 decoder ,assigning the $\overline{F_0}$ to $\overline{F_{15}}$ respectively. Then we can use Multisim to simulate the circuit.

Result

We can simulate it in the Multisim.



Discuss and comment

As the graph shown, we use 2 of 3-8 decoder to implement this function.

In the test case, when we input 0111, 1011,1101,1110, the output is always 1, but when we input other value, the input is 0. It indicates that our circuit is correct.

(We use the led light to show the output , on is 1, off is 0.)

In this experiment, I use the thought of expand the clips , the principle of standard SOP form and basic design methods. It helps me to review the knowledge and improve my ability to simulate and use Multisim.